

Case Study: NorthStar Urban Mobility and Logistics

NorthStar Urban Mobility and Logistics is a large regional organisation operating across several major UK cities. Over the last six years, it has grown quickly by merging multiple transport, delivery, and smart-city service providers into one group. It now manages public shuttle services, on-demand last-mile delivery, warehouse dispatch operations, electric vehicle charging hubs, route-planning systems, and a mobile customer platform used by passengers, drivers, dispatch managers, and commercial clients. Although the company presents itself as a fully integrated urban mobility provider, its internal systems remain fragmented, inconsistent, and increasingly difficult to manage.

Senior management has become concerned that the business is becoming less efficient despite rising demand. Complaints from passengers and business customers have increased sharply. Delays are becoming more frequent, delivery windows are being missed, vehicle downtime is rising, and the company is struggling to explain why some city zones consistently perform worse than others. The board has also noticed that operational costs have increased faster than revenue in some service areas, even though more data than ever is being collected from the company's platforms.

The organisation's core problem is that its data environment has grown faster than its ability to understand it. Older parts of the company still rely on structured operational systems originally built for separate business units. These systems record bookings, journeys, fleet schedules, staffing, charging activity, maintenance transactions, warehouse dispatch events, and customer payments in traditional tabular form. However, newer services generate more flexible and complex data that does not fit neatly into existing relational structures. These include mobile app interactions, driver event streams, route changes during active journeys, smart-sensor alerts from vehicles, customer complaint histories, service chat transcripts, GPS path logs, and platform-generated exception records when deliveries or passenger pickups fail.

At executive level, there is disagreement about what the real problem is. The operations director believes the company's difficulties are mainly caused by poor route allocation and underperforming city hubs. The customer experience director argues that service reliability is falling because the organisation is not properly connecting complaints, missed journeys, failed deliveries, and driver incidents into one view. The finance director believes that some routes and service contracts are fundamentally unprofitable, but the current reporting systems cannot reveal where losses are really occurring because cost, service, and exception data are stored in different places. Meanwhile, the technology director has warned that the company's existing database structure is no longer suitable for the volume and variety of data now being produced, especially when managers want to combine operational records with semi-structured platform and sensor data.

A recent internal review uncovered several worrying patterns, but no department could fully explain them. Some customers repeatedly report late arrival or failed service, yet their transactions appear "completed" in one system and "exception-handled" in another. In some areas, drivers are recording unusually high levels of manual route overrides, but there is no agreement on whether this reflects genuine road conditions, poor planning, or attempts to avoid performance targets. Certain charging hubs show heavy usage in platform logs but low associated utilisation in the asset monitoring reports. Some warehouses appear efficient when measured only by dispatch volume, but customer complaints and repeat delivery failures suggest hidden operational problems downstream. Managers

also suspect that some vehicle maintenance issues are being detected too late because fault events, scheduling records, and route assignments are not analysed together.

NorthStar's mobile platform has also created new challenges. Customers can book services, track vehicles, raise complaints, submit delivery instructions, rate journeys, and communicate with automated support systems. Drivers and field staff use a separate app to confirm job stages, report incidents, upload proof of completion, and log service exceptions. These interactions generate nested and highly variable data that has grown rapidly over time. Some customer cases contain long sequences of messages, repeated status changes, attachments, geolocation references, and escalation notes. Technical staff have tried flattening parts of this data for reporting, but the result has been incomplete and difficult to maintain. The company now wants a more suitable approach for storing and querying such records, especially where case histories, route exceptions, and event sequences need to be kept together.

You are expected to study the scenario, examine the available data, identify the most important operational and analytical problems, and justify your decisions. Management wants more than descriptive reporting. They want to understand where inefficiencies originate, why service quality varies across locations and service types, which factors appear to drive failure, delay, customer dissatisfaction, and cost escalation, and how the company should redesign its data environment to support better decision-making.

You are expected to work across multiple technologies because no single tool is sufficient for the task. The organisation still depends heavily on structured operational records that must be queried efficiently and combined across business functions. At the same time, the company needs analytical work that can uncover trends, relationships, anomalies, and patterns in performance. It also needs a modern NoSQL design capable of handling flexible, nested, and fast-changing operational records in a way that its current systems cannot. Senior management has therefore asked for an integrated solution in which structured data can be explored and analysed appropriately, while more complex records are redesigned into a MongoDB Atlas environment that supports operational querying, case-level tracking, and scalable analytics.

The company expects the final solution to do three things at once. First, it should reveal the underlying business problems rather than merely report symptoms. Second, it should show how structured and semi-structured data can be combined to support better analysis. Third, it should include the design and implementation of a NoSQL database on MongoDB Atlas that is genuinely useful for the organisation rather than being a purely technical exercise. The board is especially interested in whether a document-based design could support an integrated operational view of customers, journeys, exceptions, complaints, and service events without the rigidity that currently limits reporting.

Students acting as data analysts are therefore required to approach the case as if they were part of a real analytics consultancy team. They must read the scenario carefully, decide what the key organisational failures are, determine what evidence is needed, analyse the data using suitable methods, and produce justified conclusions. They are also expected to design and create an appropriate MongoDB Atlas database that reflects the organisation's operational reality. This includes deciding what kinds of records should remain relational for analytical querying, what kinds of information should be modelled as documents, how nested or evolving event histories should be

stored, and how the database design can support useful querying and performance optimisation.

This scenario is demanding because the organisation's inefficiencies arise from complex relationships across fragmented data sources and interconnected business processes. Service performance, operational control, asset utilisation, customer behaviour, and financial outcomes are influenced by multiple interacting factors rather than isolated issues. As a result, key organisational problems may only become visible through integrated analysis across different datasets and system contexts.

A strong response to this case study should demonstrate the ability to critically engage with the scenario by identifying significant analytical problems, selecting and applying appropriate analytical methods, interpreting meaningful patterns within the data, and designing a well-justified MongoDB Atlas NoSQL database solution that reflects the organisation's operational complexity and supports effective decision-making.

The NorthStar_dataset is provided via Blackboard to support data analysis, database modelling, and system optimisation activities within this case study.

Students are expected to integrate SQL (within R), R analytics, Python data processing, and MongoDB Atlas NoSQL database design as part of a unified analytical solution.

Students are required to implement their analytical workflows using Google Colab notebooks. All notebooks must be stored in a structured GitHub repository, and the repository link must be included in the final assignment submission. This requirement ensures transparency, reproducibility, and proper version control of analytical work.